Lab3-Docker

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Part1:

1. Configuring Docker Daemon:
    a. Updating the daemon.json file to trust the local insecure registry.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
GNU nano 7.2                                  /etc/docker/daemon.json
{
  "insecure-registries" : ["localhost:5000"]
}
```

2. Creating the Registry:
    a. Running the official registry container on port 5000 and restarting the Docker service.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
root@ip-172-31-78-176:~# docker run -d -p 5000:5000 --restart=always --name my-registry registry:2
Unable to find image 'registry:2' locally
2: Pulling from library/registry
6d464ea18732: Pull complete
44cf07d57ee4: Pull complete
8e82f80af0de: Pull complete
3493bf46cdec: Pull complete
bbbdd6c6894b: Pull complete
32a76c78501f: Download complete
b537bf6d1146: Download complete
Digest: sha256:a3d8aaa63ed8681a604f1dea0aa03f100d5895b6a58ace528858a7b332415373
Status: Downloaded newer image for registry:2
f6c35bff69a79e173dac9cc752bf5b86c175b3635d9828337a76ace8491cd0f7
root@ip-172-31-78-176:~# sudo nano /etc/docker/daemon.json
root@ip-172-31-78-176:~# sudo systemctl restart docker
root@ip-172-31-78-176:~#
```

3. Dockerfile for Custom Nginx:

a. Writing the Dockerfile based on alpine:latest to install and run Nginx.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
FROM alpine:latest

RUN apk add --no-cache nginx

RUN mkdir -p /run/nginx
COPY nginx.conf /etc/nginx/http.d/default.conf

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
~
~
~
~
~
~
~
```

4. Build, Tag, and Push:

a. Building the custom image, tagging it for the local registry, and pushing it.

```
root@ip-172-31-78-176:~# docker build -t my-custom-nginx .
[+] Building 1.9s (9/9) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 207B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/alpine:latest                                                          0.1s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => CACHED [1/4] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659  0.0s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659  0.0s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 181B                                                                                         0.0s
 => [2/4] RUN apk add --no-cache nginx                                                                                    0.9s
 => [3/4] RUN mkdir -p /run/nginx                                                                                         0.2s
 => [4/4] COPY nginx.conf /etc/nginx/http.d/default.conf                                                                  0.1s
 => exporting to image                                                                                                    0.4s
 => => exporting layers                                                                                                   0.2s
 => => exporting manifest sha256:bc407b22cbe5bd7fdfc1df99c0cf609ba069882a501ed7202fa326475e20f4bb                         0.0s
 => => exporting config sha256:63d84880f872d5ae6d43c4910e9955af79cb87da1141eed80b3fced6e5440b0b                           0.0s
 => => exporting attestation manifest sha256:2db33093a5f3d338a8bcb82b46fe69f79ffcd69752c2bc45b9ff3382a5c8196a             0.0s
 => => exporting manifest list sha256:97a2e6a1a091c05611a7ca5b8eeab7eccafb64b19d36dd925d339a5c19f0af73                    0.0s
 => => naming to docker.io/library/my-custom-nginx:latest                                                                 0.0s
 => => unpacking to docker.io/library/my-custom-nginx:latest                                                              0.1s
root@ip-172-31-78-176:~# docker tag my-custom-nginx localhost:5000/my-custom-nginx:v1
root@ip-172-31-78-176:~# docker push localhost:5000/my-custom-nginx:v1
The push refers to repository [localhost:5000/my-custom-nginx]
1686131391a3: Pushed
1d90ff368ba0: Pushed
589002ba0eae: Pushed
ecde18ad206d: Pushed
4f4fb700ef54: Pushed
v1: digest: sha256:97a2e6a1a091c05611a7ca5b8eeab7eccafb64b19d36dd925d339a5c19f0af73 size: 856
root@ip-172-31-78-176:~#
```

5. Testing the Registry:

a. Verifying the image exists in the local registry catalog.

```
root@ip-172-31-78-176:~# curl http://localhost:5000/v2/_catalog
{"repositories":["my-custom-nginx"]}
root@ip-172-31-78-176:~#
```

Part2:

1. Docker Compose Configuration:

a. Setting up the docker-compose.yml for WordPress and MySQL with persistent storage.

```
root@ip-172-31-78-176: ~/wordpress-lab
root@ip-172-31-78-176: ~/wordpress-lab 143x38
  GNU nano 7.2                                   docker-compose.yml *
version: '3.8'
services:
  db:
    image: mysql:5.7
    volumes:
      - ./mysql_data:/var/lib/mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: password
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wordpress
      MYSQL_PASSWORD: password

  wordpress:
    depends_on:
      - db
    image: wordpress:latest
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wordpress
      WORDPRESS_DB_PASSWORD: password
      WORDPRESS_DB_NAME: wordpress
```

2. Deployment:

a. Creating the project folder and launching the stack using docker-compose up -d.

```
root@ip-172-31-78-176: ~/wordpress-lab
root@ip-172-31-78-176: ~/wordpress-lab 143x38
root@ip-172-31-78-176:~# mkdir wordpress-lab && cd wordpress-lab
root@ip-172-31-78-176:~/wordpress-lab# nano docker-compose.yml
root@ip-172-31-78-176:~/wordpress-lab# docker-compose up -d
Command 'docker-compose' not found, but can be installed with:
snap install docker          # version 28.4.0, or
apt  install docker-compose  # version 1.29.2-6
See 'snap info docker' for additional versions.
root@ip-172-31-78-176:~/wordpress-lab# apt  install docker-compose
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-compose python3-docker python3-dockerpty python3-docopt python3-dotenv python3-texttable python3-websocket
Recommended packages:
  docker.io
The following NEW packages will be installed:
  docker-compose python3-compose python3-docker python3-dockerpty python3-docopt python3-dotenv python3-texttable python3-websock
0 upgraded, 8 newly installed, 0 to remove and 99 not upgraded.
Need to get 297 kB of archives.
After this operation, 1589 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-websocket all 1.7.0-1 [38.1 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-docker all 5.0.3-1ubuntu1.1 [89.1 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dockerpty all 0.4.1-5 [11.4 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-docopt all 0.6.2-6 [26.1 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-dotenv all 1.0.1-1 [22.3 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-texttable all 1.6.7-1 [11.0 kB]
```

Part3:

1. Image Migration:

a. Taking the image from Lab 2, tagging it, and pushing it to the new private registry.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
root@ip-172-31-78-176:~# docker tag abdelrahimbebo/iti-flask-lab2 localhost:5000/my-flask-app:v1
root@ip-172-31-78-176:~# docker push localhost:5000/my-flask-app:v1
The push refers to repository [localhost:5000/my-flask-app]
589002ba0eae: Mounted from my-custom-nginx
197f372f9582: Pushed
34fa1c34d22d: Pushed
184910d156ca: Pushed
1baa16e47791: Pushed
c09b94e1f7d9: Pushed
a99de56aa689: Pushed
v1: digest: sha256:0f80fcb32a5eeb49a889ef87caca5ba4fa654e03cd49a74e4da361309439c906 size: 856
root@ip-172-31-78-176:~#
```

2. Reverse Proxy Configuration:

a. Configuring nginx.conf to forward traffic to the Flask container.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
server {
    listen 80;
    location / {
        proxy_pass http://flask-app:5000;
    }
}
~
~
~
~
~
~
~
~
```

3. Multi-Service Compose:

a. The final Docker Compose file integrating the private Flask image and Nginx.

```
root@ip-172-31-78-176: ~
root@ip-172-31-78-176: ~ 143x38
version: '3.8'
services:
  flask-app:
    image: localhost:5000/my-flask-app:v1
    restart: always

  nginx:
    image: nginx:alpine
    ports:
      - "80:80"
    volumes:
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    depends_on:
      - flask-app
~
~
```

4. Final Execution:

a. Running the final stack to ensure both containers are up and running.

```
root@ip-172-31-78-176:~# vi nginx.conf
root@ip-172-31-78-176:~# vi docker-compose-flask.yml
root@ip-172-31-78-176:~# docker-compose -f docker-compose-flask.yml up -d
Creating network "root_default" with the default driver
Creating root_flask-app_1 ... done
Creating root_nginx_1     ... done
root@ip-172-31-78-176:~#
```